



PROJECT REPORT

NUST COLLEGE OF ELECTRICAL & MECHANICAL ENGINEERING (CEME)-MTS



Personal Info

Class: Syndicate -A

Degree: 47-Mechatronics Engineering

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Design and Analysis of RLC Multi-Stage Filter with Ripple Reduction

Introduction

This project presents the design and implementation of a multi-stage RLC filter system. The system consists of a rectifier circuit followed by an LC ripple filter to produce smooth DC voltage. A series RLC band-pass filter is then implemented to select a specific frequency from an input signal. The project demonstrates practical applications of resistor (R), inductor (L), and capacitor (C) in filtering, resonance, and ripple reduction. The complete circuit is built on a veroboard and tested using standard laboratory equipment.

Circuit Description

The circuit consists of three main stages:

Stage 1: Bridge Rectifier

Converts AC voltage into pulsating DC using four diodes.

Stage 2: LC Filter

- Inductor (10mH)
 - Capacitor (1000 μ F)
- This stage smooths the DC output and reduces 100hz ripple.

Stage 3: RLC Filter

- Resistor (470 Ω)
 - Inductor (10mH)
 - Capacitor (0.1 μ F)
- This stage demonstrates resonance behavior and high frequency filtering.

Theoretical Calculations

Rectifier Output

$$V_{\text{peak}} = \sqrt{2} \times V_{\text{rms}}$$

$$V_{\text{peak}} = 1.414 \times 12 = 16.97\text{V}$$

After diode drop:

$$V_{\text{dc}} \approx 16.97 - 1.4 = \mathbf{15.5V}$$

Ripple Voltage

Ripple Voltage = $I / (f \times C)$
= $0.015 / (100 \times 0.001)$
= **0.15V**
Final DC = **15V**

Resonant Frequency

$f_r = 1 / (2\pi\sqrt{LC})$
 $f_r = 1 / (2\pi\sqrt{0.01 \times 10^{-7}})$
 $f_r = 5032\text{Hz}$

Reactances

$X_L = 2\pi fL = 2\pi(5032)(0.01) = \mathbf{31.6\Omega}$
 $X_C = 1 / (2\pi fC) = 1 / (2\pi(5032)(10^{-7})) = \mathbf{31.6\Omega}$
 $X_L = X_C$ Resonance

Total Resistance

$R_{\text{total}} = 470 + 330 = \mathbf{800\Omega}$

Current

$I = (15 - 2) / 800$
 $I = 0.016 \text{ A} = \mathbf{16 \text{ mA}}$

Voltage Distribution (Theoretical)

- $V_{470} = 16\text{mA} \times 470\Omega = 7.52\text{V}$
- $V_{330} = 16\text{mA} \times 330\Omega = 5.28\text{V}$
- $V_L = 0.5\text{V}$
- $V_C = 0.5\text{V}$
- $V_{LED} \approx 2\text{V}$

Experimental Readings

Measured Values (Using DMM)

Point	Experimental Value
Transformer Output	20V AC
After Rectifier	23V DC
After Capacitor	20V DC
Across $1\text{k}\Omega$	20V
Inductor	1.1V
Capacitor (RLC)	19V
470Ω Resistor	5V
330Ω Resistor	8–9V

Experimental Current Calculations

470Ω Branch:

$$I = 5 / 470$$

$$I = \mathbf{10.6\text{ mA}}$$

LED Branch:

$$I = (8 - 2) / 330$$

$$I = \mathbf{18.1\text{ mA}}$$

Comparison (Theory vs Experimental)

	Theoretical	Experimental
Rectifier Output	15.5V	23V
Filter Output	15V	20V
Current	16mA	18mA
Inductor Voltage	0.5V	1.1V

Reason for Difference

The difference between theoretical and experimental values is due to:

- Transformer no-load voltage increase
- Diode and internal losses
- Component tolerances
- Non-ideal behavior of inductor and capacitor
- Load variations

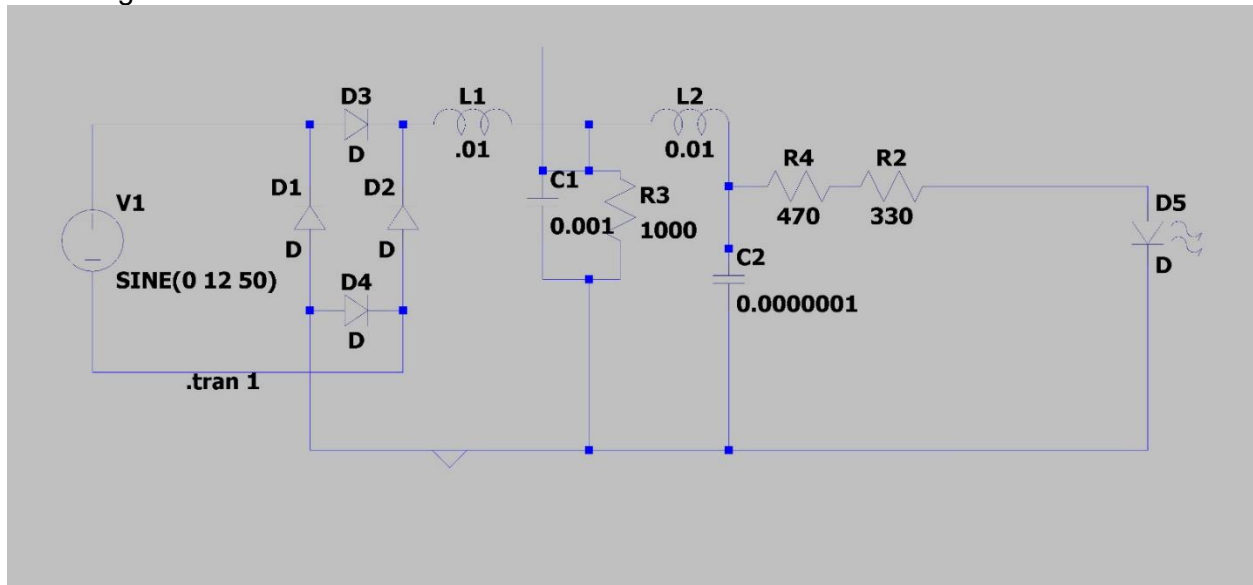
Working Explanation

The circuit works by taking high-voltage electricity from a wall outlet and transforming it into a smooth, steady flow of power used to light up an LED. Here is how it happens step-by-step:

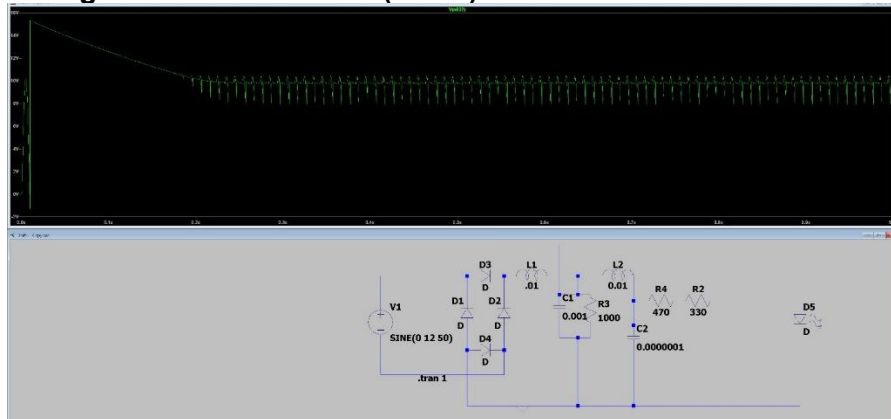
- **Step-Down Transformation:** The process starts with a transformer that lowers the high voltage from the mains to a safer level. While it is labeled as 12V, it actually measured **20V AC** in the lab because there wasn't a heavy load connected to it yet.
- **Converting AC to DC (Rectification):** This 20V AC goes into a Bridge Rectifier which is a set of four diodes. These diodes act like one way valves turning the back and forth AC into a one-way DC. After this stage the voltage measured **23V DC**.
- **Smoothing the Power (The LC Filter):** The power is still bumpy at this point, so it passes through an inductor and a large capacitor. The inductor smooths out the current, and the capacitor acts like a storage tank that fills in the gaps between pulses. This turned the bumpy 23V into a much smoother **20V DC**.
- **Filtering and Resonance (The RLC Stage):** The clean DC then enters a final stage with a resistor, another inductor, and a smaller capacitor. This part of the circuit is designed for **resonance**, a state where it becomes very efficient at handling a specific frequency which we calculated to be about **5032 Hz**. In our DC setup, it helps block out any remaining high-frequency noise.
- **Powering the LED:** Finally, the power is split into two paths. One path goes through a 470Ω resistor, and the other goes through a 330Ω resistor to light the LED. We measured about **18mA** of current flowing through the LED, which is the perfect amount to keep it bright without burning it out.

Circuit Simulation

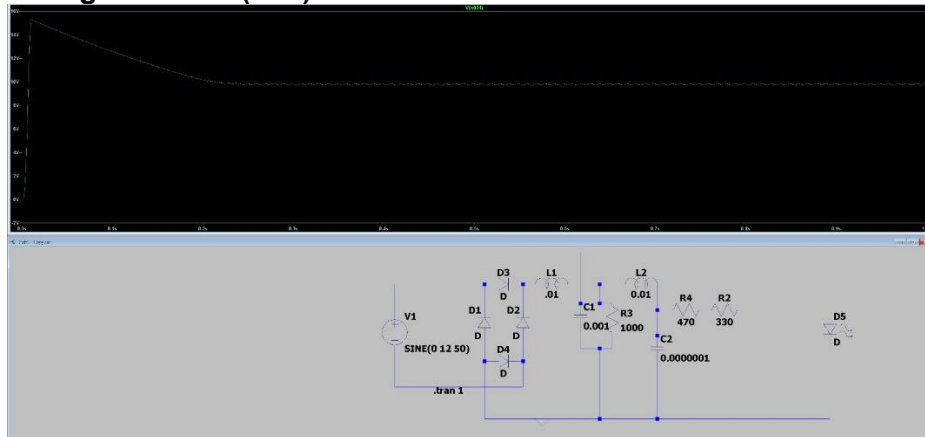
The circuit simulation was conducted using LTspice to validate the theoretical design of the multi-stage RLC filter.



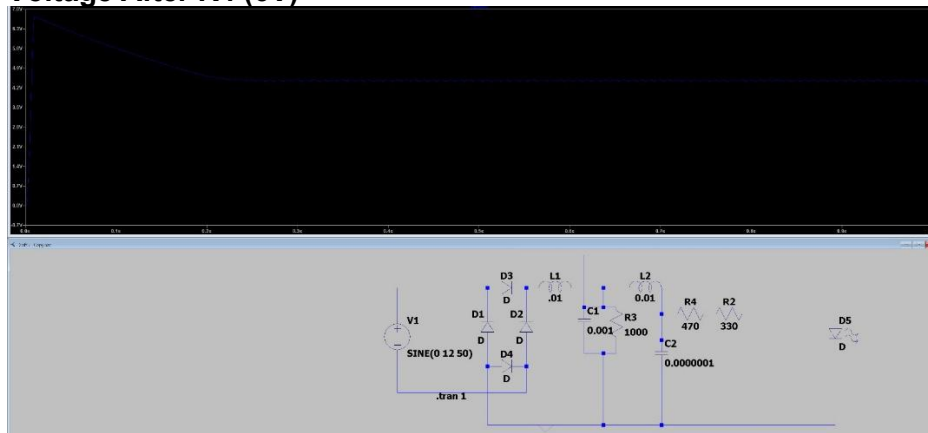
Voltage After Rectification (15.5V)



Voltage After L2 (15V)



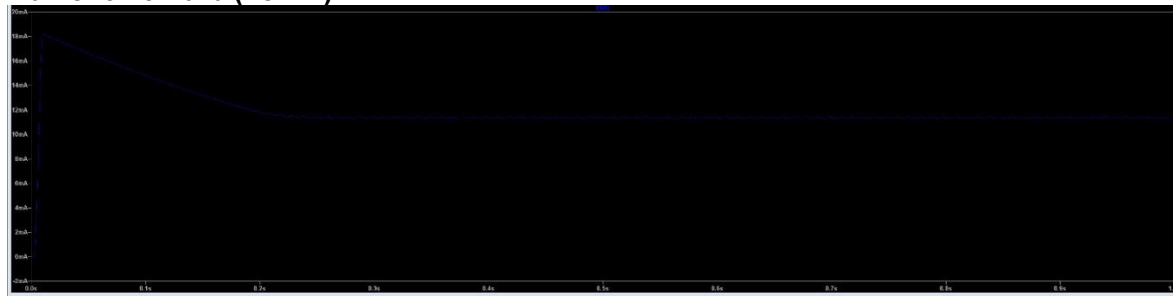
Voltage After R4 (6V)



Voltage After R2 (7.3V)



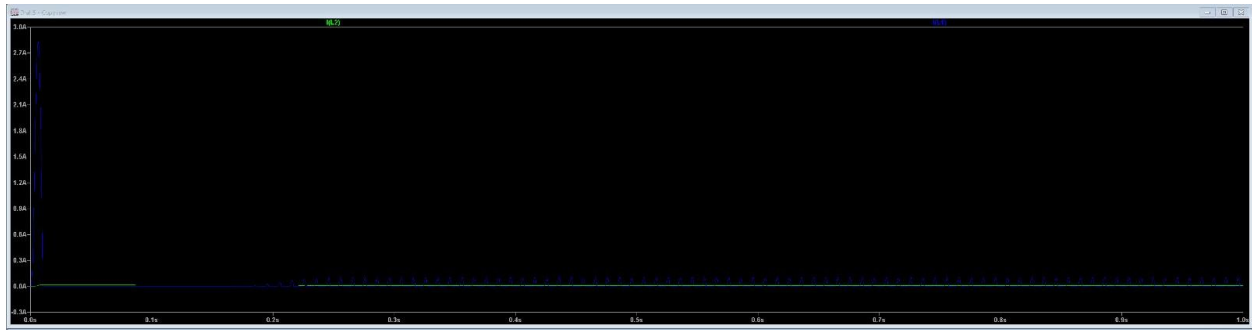
Current At Bulb (18mA)



Current At L2(Green), R4(Red), C2(Blue)



Current At L1(Blue), L2(Green)



Conclusion

The project successfully achieved the following:

- Conversion of AC to DC using a bridge rectifier
- Reduction of ripple using an LC filter
- Demonstration of resonance in an RLC circuit
- Practical verification of theoretical concepts

The experimental results closely match theoretical expectations with minor variations due to real-world conditions.